

PRACTICAL-02

AIM: Implementation and Time analysis of linear and binary search algorithm.

1.) LINEAR SEARCH

PROGRAM:

```
#include <iostream> using namespace std;

int main() { int arr[20],n,key,i;  cout <<

"Enter the number of elements: ";

    cin >> n;  cout << "Enter the array

elements: ";

    for (int i = 0; i < n; i++) {

cin >> arr[i];

    }

    cout << "Enter the element to search: ";

cn >> key;  for (int i = 0; i < n; i++) {    if

(arr[i] == key) {        cout << "Element

found at index " << i;

        return 0;

    }

}

    cout << "Element not found";

return 1;

}
```

OUTPUT:

```
Output
Enter the number of elements: 5
Enter the array elements: 9
1
6
7
2
Enter the element to search: 2
Element found at index 4

=== Code Execution Successful ===
```

2.) BINARY SEARCH

PROGRAM:

```
#include <iostream>

using namespace std;

int main()
{
    int n, key;    int
    low, high, mid;

    cout << "Enter the number of elements: ";

    cin >> n;    int a[n];    cout << "Enter the
sorted array elements: ";

    for (int i = 0; i < n; i++)
    {
        cin >>
a[i];
    }

    cout << "Enter the key element: ";

    cin >> key;    low = 0;    high = n - 1;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (a[mid] == key)
        {
```

```
    cout << "Element found at position " << mid << endl;

    break;
}

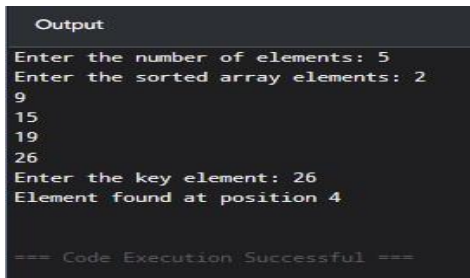
else if (a[mid] < key)
{
    low = mid + 1;
}

else
{
    high = mid - 1;
}
}

if (low > high)
{
    cout << "Element not found" << endl;
}

return 0;
}
```

OUTPUT:



```
Output
Enter the number of elements: 5
Enter the sorted array elements: 2
9
15
19
26
Enter the key element: 26
Element found at position 4

=== Code Execution Successful ===
```